Habib University
Course Code: EE252/CE251- Spring 2025
Course Title: Signals and Systems
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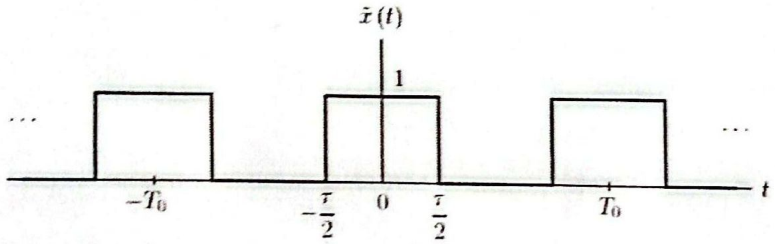
Examination: Final-II | Date: 12th April, 2025
Total Marks: 60 (15% of Course credits) | Duration: 75 minutes

CLO 2	To analyze continuous signals and systems in the frequency domain	LDL = 4 Analysis PLO=02 Problem solving

NOTE: Attempt all questions; your answer must include neat and properly labeled sketches (where applicable), a logical flow of the solutions, and detailed steps of your work.

Q1- Frequency domain analysis using Fourier Series

As per routine practice, the Engineering students of Habib University are always involved in different technical design projects starting from their first Semester courses. In one project, students worked on Pulse Width Modulated (PWM) wave generation where they understood all aspects of wave in the time domain. Now, some student wants to explore the frequency domain analysis of the PWM wave, specifically its line spectra, power, and bandwidth-related investigation. For that analysis, they consider the pulse train with **duty cycle d (here d = τ /T_0)**



$\omega_0 = 2\pi f_0$
$= \frac{2\pi}{T_0}$

I. [10] Show that the Exponential Fourier Series (EFS) coefficient **C_k = d sinc (kd)** and sketch its magnitude and phase plot for K = 0 to 10 if the duty cycle value is 0.3. What should students expect regarding the total bandwidth required for this signal in the frequency domain?

II. [10] In the first part, assume the bandwidth requirement observed is too high and students are trying to find a solution for it. Assuming the signal is processed through a lowpass filter that retains only five harmonics and eliminates all others, do you think this technique would solve the high bandwidth issue? If the total power of the signal equals 0.3, how much signal power will be preserved in five harmonics as compared to total signal power? Briefly comment on the trade-off between signal power and bandwidth in this particular investigation.

Q2 - Frequency domain analysis using Fourier Transform

Find the Fourier transform of the following signals by using the formula:

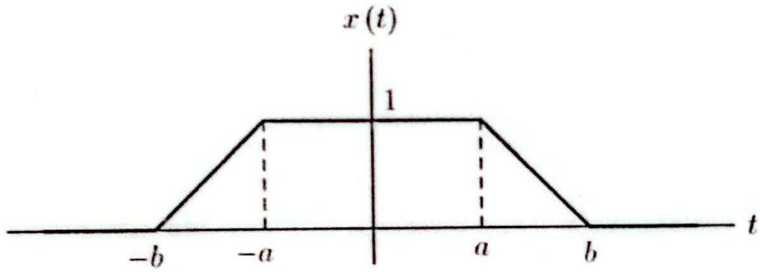
[10] $y(t) = e^{-3t}\cos(2t)\,u(t)$

[10] $x(t) = 5\,\Pi\left(\frac{t-3}{2}\right)$

Q3 - Fourier Transform properties

Using appropriate Fourier transform property, find the Fourier transform of the following signals. State the property used and briefly explain its importance in signal processing.

[10] A trapezoidal pulse x (t) for b=5 and a= 2



[10] $$z(t) = \frac{3}{1+5t^2}$$

Instructor comments do not write in this section

Q2) a). $y(t) = e^{-3t}\cos(2t)\; u(t)$

$$X(f) = \int_{0}^{\infty} e^{-3t}\cos(2t)\cdot e^{-j2\pi f t}\cdot dt$$

* $\cos(2t) = \dfrac{e^{j2t}+e^{-j2t}}{2}$

$$\int_0^\infty e^{-3t}\left(\frac{e^{j2t}+e^{-j2t}}{2}\right)\cancel{dt}\; e^{-j2\pi f t}$$

$\omega_0 = 2$

$2\pi f = 2$

$\boxed{f = 1/\pi}$

$$\int_0^\infty \left(\frac{e^{-3t+j2t}+e^{-3t-j2t}}{2}\right)\cdot e^{-j2\pi f t}\, dt$$

$$\Rightarrow \frac{1}{2}\int_0^\infty e^{-3t+j2t-j2\pi f t} + e^{-3t-j2t-j2\pi f t}\, dt$$

$$\Rightarrow \frac{1}{2}\left[\frac{e^{-3t+j2t-j2\pi f t}}{-3+j2-j2\pi f}\right]_0^\infty + \frac{1}{2}\left[\frac{e^{-(3+j2-j2\pi f)t}}{-3+-j2-j2\pi f}\right]_0^\infty$$

$$\Rightarrow \frac{1}{2}\left[\cancel{\frac{-0}{3-j2+j2\pi f}} - \frac{1}{-3+j2-j2\pi f}\right] + \frac{1}{2}\left[-\frac{1}{-3-j2-j2\pi f}\right]$$

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$$\Rightarrow \frac{1}{2}\left[\frac{1}{3-j2+j2\pi f} + \frac{1}{3+j2-j2\pi f}\right]$$

* $\dfrac{1}{2}\left[\dfrac{1}{3-(j2-j2\pi f)} + \dfrac{1}{3+(j2-j2\pi f)}\right]$

Instructor comments do not write in this section

$* \quad (3)^2 - (j2 - j2\pi f)^2$

$9 - [(j2)^2 - 2(j2)(j2\pi f) + (j2\pi f)^2]$

$9 - [-4 + 8\pi f - 4\pi^2 f^2]$

$9 + 4 - 8\pi f + 4\pi^2 f^2$

$13 - 8\pi f + 4\pi^2 f^2$

$$\Rightarrow \frac{1}{2}\left[\frac{3 + \cancel{j2} - \cancel{j2\pi f} + 3 - \cancel{j2} + \cancel{j2\pi f}}{13 - 8\pi f + 4\pi^2 f^2}\right]$$

$$\Rightarrow \boxed{\frac{3}{13 - 8\pi f + 4\pi^2 f^2}}$$

Verifying through identity known: that:

$$\int e^{ax}\cos(bx) = \frac{e^{ax}}{a^2+b^2}\left[a\cos(bx) + b\sin(bx)\right]$$

$$\int e^{-3t}\cos(2t) \Rightarrow \frac{e^{-3t}}{9+4}\left[-3\cos(2t) + 2\sin(2t)\right].$$

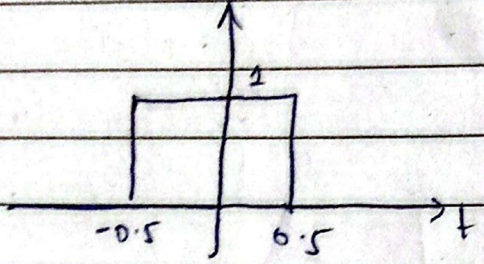
$$\Rightarrow \frac{e^{-3t}}{13}\left[-3\left[\frac{e^{j2t} + e^{-j2t}}{2}\right] + \cancel{2}\left[\frac{e^{j2t} - e^{-j2t}}{\cancel{2}j}\right]\right]$$

$\Rightarrow$

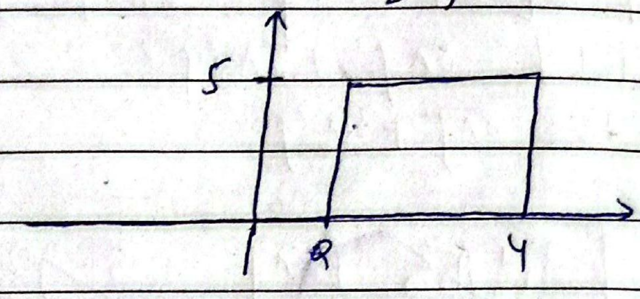
* It's more complex with this! I rest my case. :)

Q. 2)
(b) $x(t) = 5\Pi\left(\frac{t-3}{2}\right)$

$\Pi(t)$.



$5\Pi\left(\frac{t-3}{2}\right)$.



$\Rightarrow X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$

$\Rightarrow X(f) = \int_{2}^{4} 5 \cdot e^{-j2\pi ft}\, dt$.

$\Rightarrow 5\left[\frac{-e^{-j2\pi ft}}{j2\pi f}\right]_2^4$.

$\Rightarrow 5\left[\frac{-e^{-j8\pi f}}{j2\pi f} + \frac{e^{-j4\pi f}}{j2\pi f}\right]$.

$\Rightarrow \frac{5}{\pi f}\left[\frac{e^{-j4\pi f} - e^{-j8\pi f}}{2j}\right]$

$= \frac{5}{\pi f}\left[e^{-j6\pi f}\left(\frac{e^{j2\pi f} - e^{-j2\pi f}}{2j}\right)\right]$.

$\Rightarrow \frac{5e^{-j6\pi f}}{\pi f} \sin(2\pi f)$.

$\Rightarrow \frac{10e^{-j6\pi f}}{2\pi f} \sin(2\pi f)$

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$\boxed{X(f) \Rightarrow 10e^{-j6\pi f}\, \mathrm{sinc}(2f)}$.

Instructor comments do not write in this section

(Q. 3) (a) $(-b, 0)\ (-a, 1)$ for $-b < t < a$

$\hookrightarrow m = \dfrac{1-0}{-a+b} = \dfrac{1}{b-a}$

$0 = \dfrac{1}{b-a}(-b) + c$

$c = \dfrac{b}{b-a} \Rightarrow \boxed{y = \dfrac{t+b}{b-a}}$

$\boxed{1 \text{ for } -a < t < a}$

$(a, 1)\ (b, 0)$ for $a < t < b$.

$m = \dfrac{1-0}{a-b} = \dfrac{-1}{b-a}$

$0 = \dfrac{-1}{b-a}(b) + c$

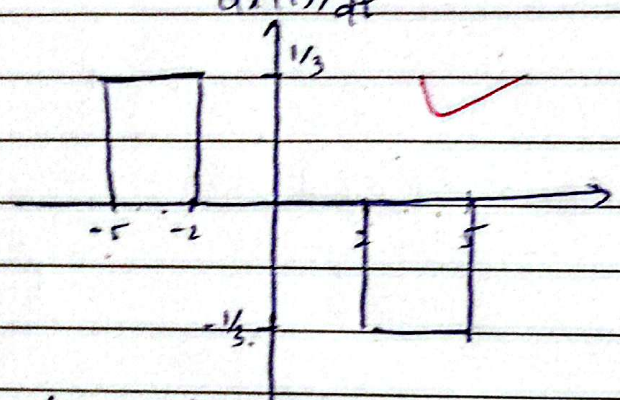
$c = \dfrac{b}{b-a} \Rightarrow \boxed{y = \dfrac{-t+b}{b-a}}$

$$x(t) = \begin{cases} \dfrac{t+b}{b-a} & \text{for } -b \leq t < a \\ 1 & \text{for } -a < t < a \\ \dfrac{-t+b}{b-a} & \text{for } a < t < b. \end{cases}$$

~~$x(t) = \begin{cases} \frac{5}{3} = 2 & \text{for } -5 < t < -2. \\ 1 & \text{for } -2 < t < 2 \\ \frac{4}{2} = 2 & \text{for} \end{cases}$~~

differentiate to make it better.

$$\frac{dx(t)}{dt} = \begin{cases} \frac{1}{b-a} & \text{for } -b < t < -a \\ \frac{-1}{b-a} & \text{for } a < t < b. \\ 0 & , \text{otherwise.} \end{cases}$$



$$x(t) = \frac{1}{3}\Pi\left(\frac{t+3.5}{3}\right) - \frac{1}{3}\Pi\left(\frac{t-3.5}{3}\right).$$

$$x(t)^3 = \frac{1}{3}\Pi\left(\frac{t}{3}+\frac{7}{6}\right) - \frac{1}{3}\Pi\left(\frac{t}{3}-\frac{7}{6}\right)$$

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$$X(\omega) = \frac{3e^{+j\omega 7/6}}{3}\,\text{sinc}\left(\frac{3\omega}{2\pi}\right) - \frac{3e^{-j\omega 7/6}}{3}\,\text{sinc}\left(\frac{3\omega}{2\pi}\right).$$

$$X(f) = \text{sinc}(3f)\left[e^{j\frac{7}{3}\pi f} - e^{-j\frac{7}{3}\pi f}\right]$$

$$e^{\pi + \frac{7}{3}f}$$

$$= \text{sinc}(3f)\left[e^{j\pi}\cdot e^{7/3 f} - e^{-j\pi}e^{7/3 f}\right].$$

$$e^{\pm j\pi} = -1$$

$$=$$

$$\boxed{X(f) = \text{sinc}(3f)\left[e^{7/3 f} - e^{-7/3 f}\right].}$$

Instructor comments do not write in this section

Q. 3) (b) $x(t) = \dfrac{3}{1+5t^2}$

Suppose:

$x(t) = e^{-a|t|} \xleftrightarrow{\mathcal{F}} \dfrac{2a}{a^2+\omega^2}$

* Using duality we can say that.

$\dfrac{2a}{a^2+t^2} \xleftrightarrow{\mathcal{F}} 2\pi e^{-a|-\omega|}$

* Since, $e^{-a|-\omega|}$ is an even function. Hence,

$e^{-a|-\omega|} = e^{-a|\omega|}$

$\dfrac{2a}{a^2+t^2} \xleftrightarrow{\mathcal{F}} 2\pi e^{-a|\omega|}$

$\dfrac{10a}{5a^2+5t^2} \xleftrightarrow{\mathcal{F}} 2\pi e^{-a|\omega|}$

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* $5a^2 = 1 \Rightarrow a = 1/\sqrt{5}$

$\dfrac{2\sqrt{5}}{1+5t^2} \xleftrightarrow{\mathcal{F}} 2\pi e^{-|\omega|/\sqrt{5}}$

* Multiply by $3/2\sqrt{5}$

$\dfrac{3}{\cancel{2\sqrt{5}}} \dfrac{\cancel{2\sqrt{5}}}{1+5t^2} \xleftrightarrow{\mathcal{F}} \dfrac{3}{\cancel{2}\sqrt{5}} \cancel{2}\pi e^{-|\omega|/\sqrt{5}}$

$\boxed{\dfrac{3}{1+5t^2} \xleftrightarrow{\mathcal{F}} \dfrac{3\pi}{\sqrt{5}} e^{-|\omega|/\sqrt{5}}}$

$\dfrac{3}{2\sqrt{5}} \cdot 2\sqrt{5} = 3$

* Fourier Transform is quite complex so when we use FT we know if we do inverse it will be the same and we can get back to our signal

* Duality helps in going back to its signal either in Frequency domain or Time domain for analysis.

Q.1) II).

first five harmonics of $|C_x|$'s power.

$$(0.3)^2 + \left[(0.295)^2 + (0.282)^2 + (0.261)^2 + (0.233)^2\right] \times 2 = \boxed{0.3795}$$

should be less

⇒ retaining low bandwidth would require more power as compared to total signal power. since, power needed for five harmonics is 0.3795 while 0.3 for total signal power with too high bandwidth.

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Instructor comments do not write in this section

Q.1) I)

$$c_k = \frac{1}{2T_0}\int_{-\tau/2}^{\tau/2} 1 \cdot e^{-jk\omega_0 t}\, dt \qquad \omega_0 = \frac{2\pi}{2T_0}$$

$$\omega_0 = \frac{\pi}{T_0}$$

$$c_k = \frac{1}{2T_0}\left[\frac{e^{-jk\omega_0 t}}{-jk\omega_0}\right]_{-\tau/2}^{\tau/2}$$

$$\cancel{= \frac{1}{2T_0}\left[e^{-jk\left(\frac{2\pi}{T_0}\right)t}\right.}$$

$$= \frac{1}{2T_0}\left[\frac{-e^{-jk\omega_0\tau/2}}{jk\omega_0} + \frac{e^{jk\omega_0\tau/2}}{jk\omega_0}\right].$$

$$\Rightarrow \frac{1}{T_0 k\omega_0}\left[\frac{e^{jk\omega_0\tau/2} - e^{-jk\omega_0\tau/2}}{2j}\right]$$

$$\Rightarrow \frac{1}{T_0 k\omega_0}\sin\left(\frac{k\omega_0\tau}{2}\right)$$

$$\cancel{\Rightarrow \frac{1}{T_0}\,\frac{\sin\left(\frac{k\omega_0\tau}{2}\right)}{k\omega_0}}$$

$$\Rightarrow \frac{\tau}{2T_0}\,\frac{\sin(k\omega_0\tau/2)}{\frac{k\omega_0\tau}{2}} \Rightarrow \cancel{\frac{\tau}{2T_0}\sin c\left(\frac{k\omega_0\tau}{2}\right)}.$$

$$\Rightarrow$$

$$\cancel{\frac{k\frac{2\pi}{T_0}\tau}{2} \Rightarrow \frac{k\pi\tau}{T_0} \Rightarrow k\pi d}$$

$$\cancel{\Rightarrow \frac{\tau}{2T_0}\sin\left(\frac{k\omega_0\tau}{2}\right)}$$

$$* \quad \frac{k\frac{2\pi}{T_0}\tau}{2} = \frac{k\pi\tau}{T_0} = k\pi d$$

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$$\Rightarrow \frac{1}{T_0 k \omega_0} \sin\left(\frac{k\omega_0 \tau}{2}\right)$$

$* \ \omega_0 = 2\pi / T_0$.

$$\Rightarrow \frac{1}{\cancel{T_0}\left(\frac{2\pi}{\cancel{T_0}}\right)k} \sin\left(\frac{k\frac{2\pi}{T_0}\tau}{2}\right) \Rightarrow \frac{1}{2\pi k} \sin\left(\frac{k\pi\tau}{T_0}\right).$$

$$\Rightarrow \frac{1}{\pi k} \sin\left(k\frac{\pi}{2}d\right).$$

τ ?
Steps missing?

$$\Rightarrow \frac{d}{} \frac{\sin(k\pi d)}{k\pi d} \Rightarrow \frac{d}{} \operatorname{sinc}(kd). = C_k.$$

Magnitude and Phase.

$c_k = d \operatorname{sinc}(kd)$.

$c_k = \frac{\cancel{d} \sin(kd)}{k\cancel{d}} = \frac{\sin(kd)}{k} \Rightarrow \frac{\sin(0.3k)}{k}$.

k	$\lvert c_k \rvert$
0	0.3.
1	0.295
2	0.2823
3	0.261
4	0.233
5	0.199
6	0.162
7	0.123
8	0.084
9	0.047
10	0.0141

Not correct.

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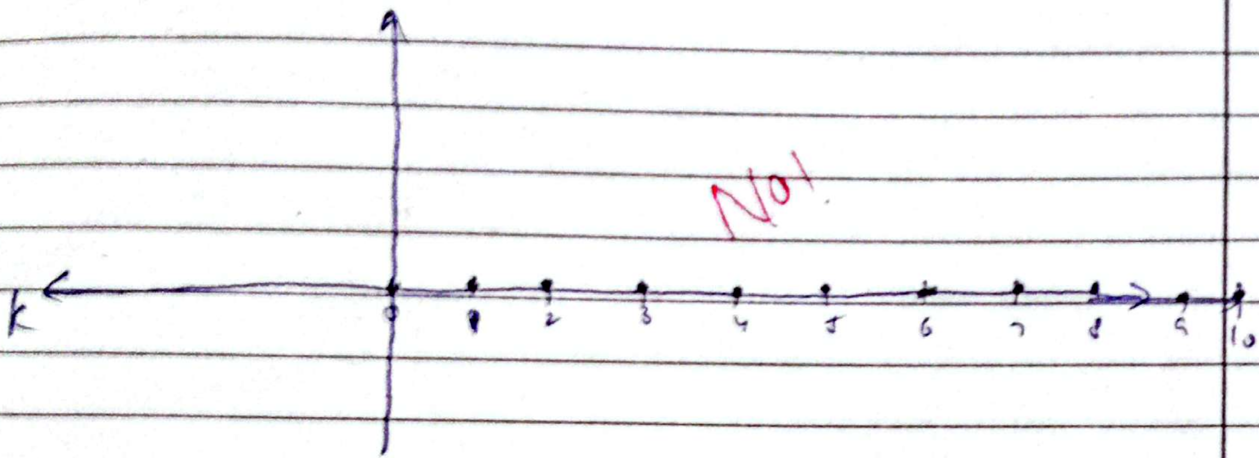
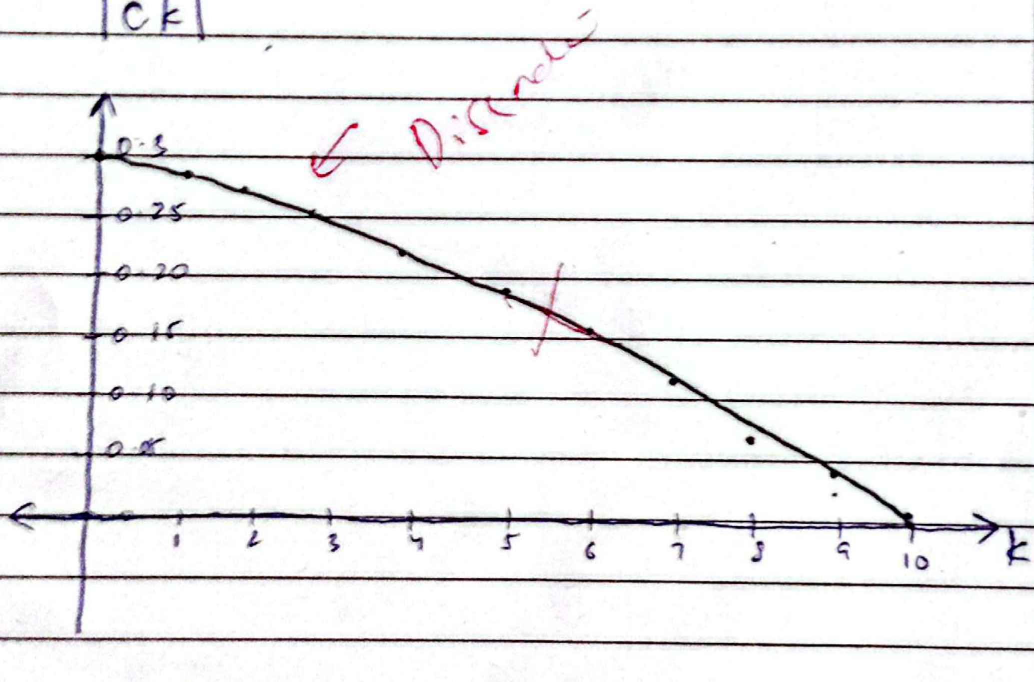
$\Rightarrow \dfrac{\sin(0.3k)}{k} =$

$\angle c_k$



$|c_k|$



+ Students should expect a sinc function that decays over time as it goes to ∞ from 0 and $-\infty$ from 0.

Bandwidth $\rightarrow \alpha$